Dr. Zhu Li
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Director, NSF I/UCRC Center for Big Learning at UMKC
Associate Editor-in-Chief, *IEEE Trans on Circuits & System for Video Technology*
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Education

- **Doctor of Philosophy** in Electrical & Computer Engineering (2004)
  Northwestern University, Evanston, Illinois 60208, USA
- **Master of Science** in Computer Science (1997)
  University of Louisiana, Lafayette, LA 70504, USA
- **Bachelor of Science** in Computer Science (1992)
  Sichuan University, Chengdu, China

Research Interests

1. **Immersive Visual Media Compression and Communication**
   a. Point Cloud Capture and Compression, Graph Signal Processing.
   b. Low latency point cloud communication system and services for auto driving
   c. Light field compression and sub-aperture image super-resolution
   d. Immersive visual communication system research, source channel coding, low latency transport
2. **Future Video Coding**
   a. Subspace methods in new intra coding schemes
   b. Deep learning intra coding, chorma prediction and frame interpolation
   c. Deep learning deblocking and super resolution in coding
   d. Learning based end to end compression
3. **Machine Learning and Image Understanding**
   a. Deep Learning model compression and acceleration. Embedded system friendly deep learning capabilities for visual signal compression, recognition and re-identification.
   b. Robust object re-identification against large scale data base, new key point features with depth info, hyper-spectral imaging, high efficiency local feature aggregation and hashing. Differential key points deep learning for low resolution/quality image recognition
   c. Hyperspectral image classification, super-resolution and compression for tasks
   d. Computer vision under adverse imaging conditions. Low light, low resolution, compressed domain, and RF domain vision problems.

Prior Experiences

1. *AFRL Summer Faculty*, UAV Research Center, **US Air Force Academy**, 2016-18, 2020, 2022-23
2. Summer Faculty (1/2 time), **Air Force Research Lab**, Sensor Directorate, Dayton Ohio, 2021.
3. *Senior Staff Researcher/Sr. Manager*, Multimedia Standards Lab, **Samsung Research America**, Richardson, TX, USA, 2012-2015
4. *Senior Staff Researcher*,Media Lab, **FutureWei Technology**, Bridgewater, NJ, USA, 2010-2012
5. *Assistant Professor*, Dept of Computing, **The Hong Kong Polytechnic University**, 2008-2010
6. *Senior, Senior Staff and Principal Staff Research Engineer*, Multimedia Research Lab, **Motorola Labs**, Schaumburg, IL, USA, 2000-2008

Teaching Performance

- Overall (3 Years Average) Teaching Evaluation Score: **4.23/5.00**
- Covering image, video and multimedia computing and communication topics area.
- **Developed four new courses at UMKC**.
  - ECE 484 Digital Image Processing (A Modern Approach)
  - ECE/CS 5582 Computer Vision
  - ECE/CS 5578 Multimedia Communication

- ECE/CS 5590 Advanced Multimedia Communication

Student Advising/ Mentoring Records

- Current Students at UMKC:
  - **Currently I have 10 PhD and 3 MS Students**
    - 1 of my PhD Students Received the UMKC School of Graduate Studies (SGS) Research Grant Award.
    - 1 of my MS student nominated for UMKC uutstanding MS thesis award 2017
    - 6 of my PhD students received internships from leading tech companies: Qualcomm, HERE, Tencent Media Lab, INTEL, PolyCom, Dolby Lab, SONY, FutureWei Media Lab.
- Supervised Post-Docs:
  - Dr. Li Li, 2016.10-2020.09, now Associate Professor with Univ of Sicence & Tech of China (USTC).
  - Dr. Renlong Hang, 2018.09-2019.12, now Assistant Professor with Nanjing Univ of Info Science & Tech (NUIST).
- Graduated PhD student:
  - Dr. Yangfan Sun, 2023, initial job: Amazon Research, Shenzhen.
  - Dr. Anique Akhtar, 2022, initial job: Qualcomm Video Codec researcher.
  - Dr. Zhaobin Zhang, 2019 (initial employment: Researcher at ByteDance, San Diego)
  - Dr. Dewan F. Noor, 2019 (initial employment: Asst Prof, Tuskegee University)
  - Dr. Wei Jia, 2021, (initial employment: Researcher at Kwai's Video Codec Group, San Diego)
- Supervised CSC visiting PhD Students:
  - Yue Li, USTC, 2017-18, now with ByteDance, San Diego
  - Wenjie Zhu, Shanghai Jiaotong Univ, 2017~18
  - Han Zhang, Shanghai Jiaotong Univ, 2018~19
  - Yixin Mei, Xian Jiaotong Univ, 2019-20
- Graduated MS Thesis Students at UMKC: 3 (2 *Employed by Honeywell, American Express*)

Awards and Honors

1. *Senior Member,* Institute of Electrical and Electronic Engineers (IEEE)
2. *Distinguished Lecturer,* APSIPA, 2016-2017.
3. *Best Paper Award*, IEEE Int'l Conference on Multimedia & Expo, Toronto, 2006
4. *Best Paper Award*, IEEE Int'l Conference on Image Processing, San Antonio, 2007
5. *Best Paper Award*, IEEE Int'l Symposium on Technologies for Homeland Security, Woburn, MA, 2018.
6. Air Force Research Lab *Summer Faculty Fellowship*, 2016~18, 2020, 2022~23.

Active Grants and Pending Proposals

Active Grants:

1. *Title:* **Phase 2 Neuro Compaction for SAR Image Compression**
   Agency: AFRL STTR Phase 2; *Approved Budget:* $1.2M; *Duration:* 2 Years; 2023-2025
   Dr. Li's Role (Effort): UMKC PI for AtomBeam and UMKC joint project, UMKC portion: $496K
2. *Title:* **Phase 2 Lossless Neuro Compaction for Sensor and Control Data**
   Agency: Space Force STTR Phase 2; *Approved Budget:* $1M; *Duration:* 2 Years; 2023-2025
   Dr. Li's Role (Effort): UMKC PI for AtomBeam and UMKC joint project, UMKC portion: $288K
3. *Title:* **Phase I I/UCRC University of Missouri-Kansas City: Center for Big Learning**
   Agency: NSF; *Approved Budget:* $750K; *Duration:* 5 Years; 2018-2023
   Dr. Li's Role (Effort): UMKC PI and Overall Project Coordinator.
4. *Title:* **RINGS: L-RIM: Learning based Resilient Immersive Media Compression, Delivery and Interaction**
   Agency: NSF ; *Approved Budget:* $996K; *Duration:* 3 Years; 2022.06-2025.06
   Dr. Li's Role (Effort): UMKC Co-PI ($496K), PI: Prof. Shiwen Mao, Auburn Univ.
5. *Title:* **NSF CBL Project - Deep Learning in Future Video Compression**
   Agency: NSF/Qualcomm; *Approved Budget:* $250K; *Duration:* 3 Years; 2020.11-2023.11.

Dr. Li's Role (Effort): UMKC PI (100%)

6. *Title:* **Equity-Forward Workforce Development Pipeline for Naval STEM Superiority**
   Agency: ONR; *Approved Budget:* $758K; *Duration:* 3 Years; 2022.01-2024.12.
   Dr. Li's Role (Effort): UMKC Co-PI, PI: Prof. Dan McIntosh.
7. Title: **Unlocking the Mysteries of a Medieval Chant Book with Multispectral Imaging**
   Agency: National Endowment for Humanity (NEH)
   Requested Budget: $324K, *Duration:* 36 Months; PI, Jeff Rydberg-Cox PI, *Dr. Li's Role and Effort:* Co-PI (25%)

Completed Grants:

8. *Title:* **Phase 1 STTR Neuro Compaction of SAR and LiDAR Data**
   Agency: AFRL; *Approved Budget:* $70K; *Duration:* 0.5 Years; 2022.10-2023.04.
   Dr. Li's Role (Effort): UMKC PI (100%), outcome: AFRL invited us to go for Phase 2 1.2M STTR
9. *Title:* **Dynamic, Data-Driven Design Methodologies for IoT-Based Computer Vision Systems**
   Agency: AFOSR, *Budget:* $933K; Duration: 4years; 2018-2021
   Dr. Li's Role (Effort): Co-PI (44%, $414K), PI Prof. Shuvra Bhattacharyya, Maryland.
10. *Title:* **NSF Center for Big Learning (CBL) Project - Deep Learning for Future Video Coding**
    Agency: NSF CBL; *Approved Budget:* $50K; *Duration:* 1 Years; 2018.06-2019.06
    Dr. Li's Role (Effort): UMKC PI (100%)
11. *Title:* **NSF CBL Project - Video Deduplication**
    Agency: Inno Peak Tech; *Approved Budget:* $50K; *Duration:* 1 Years; 2021.09-2022.08.
    Dr. Li's Role (Effort): UMKC PI (100%)
12. *Title:* **Supplement Award for AFRL Research Internship in EO and SAR Sensor Fusion and Target Recognition**
    Agency: NSF; *Approved Budget:* $110K; *Duration:* 1 Years; 2022.01-2022.12.
    Dr. Li's Role (Effort): UMKC PI (100%)
13. *Title:* **NSF Center for Big Learning (CBL) Project - Mobile Edge Depth Sensing and Point Cloud Computing Services**
    Agency: NSF CBL; *Approved Budget:* $50K; *Duration:* 1 Years; 2018.06-2019.06
    Dr. Li's Role (Effort): UMKC PI (100%)
14. *Title:* **NSF Center for Big Learning (CBL) Project - Low Resolution and Quality Image Understanding**
    Agency: NSF CBL; *Approved Budget:* $50K; *Duration:* 1 Years; 2018.06-2019.06
    Dr. Li's Role (Effort): UMKC PI (100%)
15. *Title:* **University of Missouri-Kansas City Planning Grant: I/UCRC for Big Learning**
    Agency: NSF; *Budget Allocated:* $15K; *Duration:* 12 Months; 2017.02-2018.02
    Dr. Li's Role (Effort): PI and Overall Project Coordinator (50%)
16. *Title:* **Mobile Human Re-Identification**
    Agency: EyeVerify, *Type of Funding:* Contractual Research
    Approved Budget: $58K, *Duration:* 1 Years (2016.08-2018.08),
    Dr. Li's Role (Effort): PI (50%), Co-PI: Dr. Reza Derakhashani
17. *Title:* **Single and Multi-Frame Deep Learning Super-Resolution**
    Agency: EyeVerify, *Type of Funding:* Contractual Research
    Approved Budget: $77K, *Duration:* 1 Years (2016.08-2018.08),
    Dr. Li's Role (Effort): Co-PI (50%), PI: Dr. Reza Derakhashani
18. *Title:* **Future Video Coding Research**
    Agency: FutureWei Technology, *Type of Funding:* Gift Grant
    Approved Budget: $150K, *Duration:* 1.5 Years (2017.07-2018.10),
    Dr. Li's Role (Effort): PI (100%)
19. *Title:* **Deep Learning in Image Compression**
    *Agency:*Snapchat, *Type of Funding:* Gift Grant (featured in UMKC gift foundation news)
    Approved Budget: $10K, *Duration:* 1 Years (2017.07-2018.07),
    Dr. Li's Role (Effort): PI (100%)
20. *Title:* **Machine Learning in Video Transcoding**
    Agency: FutureWei Technology, *Type of Funding:* Gift Grant

Approved Budget: $95K, *Duration:* 1 Years (2017.08-2018.08),
Dr. Li's Role (Effort): PI (100%)

21. *Title:* **Low Light Image Enhancement**
    *Agency:*Snapchat, *Type of Funding:* Gift Grant (featured in the UMKC gift foundation news)
    Approved Budget: $20K, *Duration:* 1 Years (2018.09-2019.08),
    Dr. Li's Role (Effort): PI (100%)
22. *Title:* **Explore Immersive Visual Communication**
    Agency: Qualcomm, *Type of Funding:* Gift Grant bbe
    Approved Budget: $50K, *Duration:* 1 Years (2017.01-2017.12),
    Dr. Li's Role (Effort): PI (100%)
23. *Title:* **Point Cloud Compression**
    Agency: Qualcomm, *Type of Funding:* Contractual Research
    Approved Budget: $50K, *Duration:* 1 Years (2016.01-2016.12),
    Dr. Li's Role (Effort): PI (100%)
24. *Title:* **Synthetic Visual Content Coding**
    Agency: Samsung, *Type of Funding:* Gift Grant
    Approved Budget: $30K, *Duration:* 1 Years (2016.05-2017.05),
    Dr. Li's Role (Effort): PI (100%)

Proposals Under Review:

1. Title: **Rotary and Fixed-Wing Aircraft Landing Zone Identification from 3D Data Analytics (Manned or Unmanned)**
   Agency: AWERX STTR Direct Phase II
   Requested Budget: $950K, *Duration:* 24 Months; PI, Matt Coleman, Pointerra, *Dr. Li's Role and Effort:* Co-PI ($250K)

Professional and Synergistic Activities

Associate Editor-in-Chief (2020~2023): IEEE Transactions on Circuits & System for Video Technology (T-CSVT)

Associate Editor (2019-): IEEE Trans on Image Processing (T-IP)

Associate Editor (2015 - 2018): IEEE Transactions on Multimedia (T-MM)

Associate Editor (2016 - 2019): IEEE Transactions on Circuits & System for Video Technology (T-CSVT)

Associate Editor (**2020**-): Frontier in Signal Processing - Image Communication.

Associate Editor (2015 - 2018): Journal of Signal Processing Systems (JSPS), Springer Publication

Guest Editor:

- IEEE Journal on Selected Areas in Communication (J-SAC), special issue on Economics & Optimization in Networks, 2018.
- IEEE Trans on Circuits & System for Video Tech, special issue on Point Cloud Processing & Compression, 2020.

Elected Technical Committee Member:

- IEEE Multimedia Signal Processing Technical Committee (MMSP-TC): 2015-2019.
- IEEE Image, Video and Multidimensional Signal Processing (IVMSP) Technical Committee: 2018-current
- IEEE Multimedia Communication Technical Committee (MMTC): Vice Chair Asia (2008-2010), Standards Liaison (2015-2016), Steering Chair (2017-2019).
- IEEE Visual Signal Processing and Communication Technical Committee (VSPC-TC) : 2017-current
- IEEE Multimedia System and Applications Technical Committee (MSA-TC): 2016-current.

Steering Committee:

- IEEE Int'l Conference on Multimedia & Expo (ICME), 2018-2019, representing SPS, 2016-2017 representing ComSoc.
- IEEE Int'l Conference on Visual Communication & Image Processing (VCIP), 2018-2020.

Conference Chair:

- Program Co-Chair, IEEE Multimedia Signal Processing Workshop (MMSP), Poitiers, France, 2023.
- General Co-Chair, IEEE Int'l Conf on Multimedia & Expo (ICME), Brisbane, Australia, 2023.
- TPC Chair (Lead): *IEEE Visual Communication & Image Processing* (VCIP), Suzhou, China, 2022
- Elected TPC Co-Chair, representing IEEE SPS, *IEEE Int'l Conf on Multimedia & Expo* (ICME), 2019.
- General Chair, *IEEE Visual Communication & Image Processing* (VCIP), St. Petersberg, FL, USA, 2017.
- TPC Co-Chair, The 7th Annual IEEE Consumer Communications and Networking Conference (CCNC), Las Vegas, USA, 2009.

Symposium Chair:

- IEEE Global Communications Conference: Communications Software, Services and Multimedia Apps Symposium, 2017.
- ITC 28, 2016.
- IEEE International Conf on Communication (ICC) - Multimedia Services, Communication Software and Services Symposium, 2010.

Best Paper Awards Committees:

- IEEE International Conf on Multimedia & Expo, 2010.
- IEEE Trans on Circuits & System for Video Tech (T-CSVT), 2020.

Area Chair:

- IEEE International Conf on Image Processing (ICIP), 2015, 2016, 2017, 2018, 2019, 2020, 2021.
- IEEE International Conf on Multimedia & Expo (ICME), 2015, 2016, 2017, 2018, 2021.
- IEEE International Conf on Audio, Speech & Signal Processing (ICASSP), 2020, 2021
- IEEE Visual Communication and Image Processing (VCIP) Conf, 2020, 2021
- ACM Multimedia (MM), 2017, 2018.
- ACM Multimedia Systems (MMSys), 2018.
- 28th International Teletraffic Congress (ITC), 2016.

Conference Track Chair:

1. Lecture session 5: Denoising, filtering and analysis, IEEE Multimedia Signal Processing Workshop (MMSP), London, UK, 2017
2. MA.L8: Indexing & Retrieval, IEEE International Conf on Image Processing (ICIP), Phoenix, USA, 2016.
3. MP.L8: Indexing and Retrieval I, IEEE International Conf on Image Processing (ICIP), Hong Kong, 2010.

Special Session Organizer:

1. Z. Li and G. Cheung, Special Sessions Co-Chair, IEEE International Conference on Multimedia & Expo, Hong Kong, 2017.
2. Z. Li and A. K. Katsaggelos, "Content Aware Video Coding/Communication for Ubiquitous Adaptation", special session for IEEE Int'l workshop on Multimedia Signal Processing (MMSP), 2005.
3. Z. Li, H. Wang, and J. Huang, "Intelligent Video Communication: Collaboration and Adaptation", special session for 16th Packet Video workshop (PV), 2006.
4. Z. Li, P. Frossard, and A. K. Katsaggelos, "Multi-User Collaboration and Cross Layer Optimization in Wireless Multimedia Communication", special session for IEEE Int'l Conf. on Multimedia and Expo (ICME), 2006.
5. Z. Li, Y. Fu, and S. Yan, Video Analysis & Event Recognition, MMM 2010.

Program Committee Member:

1. IEEE Int'l Symposium on Multimedia (ISM), Irvine, California, 2005, 2006
2. IEEE GLOBECOM, 2006, 2007, 2008, 2010, 2011, 2012, 2013, 2014.
3. IEEE Int'l Conference on Communication (ICC), 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015.
4. PacketVideo Workshop, 2006, 2009, 2010, 2012, 2013.
5. IEEE Int'l Conference on Multimedia and Expo (ICME), 2006, 2007, 2008, 2010, 2013, 2014.
6. IEEE Consumer Communications and Networking Conference (CCNC), 2007, 2008.
7. IEEE Wireless Communication & Networking Conference (WCNC), 2013, 2014.

8. Pacific Conference on Multimedia (PCM), 2007, 2009.
9. IEEE Multimedia Signal Processing (MMSP) workshop, 2006, 2007, 2008, 2011.
10. IEEE Embedded Computer Vision Workshop, in conjunction with ICCV, 2009, 2013.
11. IEEE Workshop on Video-oriented Object and Event Classification, in conjunction with ICCV, 2009, 2010, 2011.
12. 6th IEEE Int'l Conf on Advanced Video and Signal Based Surveillance (AVSS), Genoa, Italy, 2009.

Conference Session Chair: Chaired more than twenty (20) Sessions in Various IEEE and ACM Conferences

Professional Membership: IEEE Senior Member since 2007, nominated for IEEE Fellow by CASS, 2020.

Funding Agency Services:

1. NSF Panel, 2018, 2021, 2022
2. NSF workshop on Future of the networked AR/VR, 2018.
3. Invited CAREER Award reviewer, 2018.
4. AFOSR YIP Award reviewer, 2018.
5. Reviewer for Swiss NSF, 2021
6. RGC Panelist for Hong Kong Research Grant Council, 2022, 2023

Journal Review Services:

1. IEEE Trans. on Circuits & System for Video Technology (T-CSVT)
2. IEEE Trans. on Multimedia, (T-MM)
3. IEEE Trans. on Pattern Analysis & Machine Intelligence (T-PAMI)
4. IEEE Trans. on Image Processing (T-IP)
5. IEEE Trans. on Networking (T-NET)
6. IEEE Trans. on Communication (T-COM)
7. IEEE Trans. on Information Forensic and Security (T-IFS)
8. IEEE Trans. on Parallel and Distributed Systems (T-PDS)
9. IEEE Journal on Selected Area in Communication (J-SAC)
10. IEEE Communication Magazine
11. IEEE Trans on Broadcasting (ToB)
12. ACM Trans on Internet Tech
13. ACM Trans on Multimedia Computing, Communication and Applications (TOMM)
14. Springer Journal of Multimedia Systems
15. Neurocomputing
16. EURASIP Journal on Applied Signal Processing
17. Journal of Visual Communication and Image Representation. (J-VCIR)
18. Journal of Electronic Imaging (J-EI)

Publications

Book Chapters:

1. L. Li and **Z. Li**, "Light Field and Plenoptic Point Cloud Compression", Springer: *Handbook for Dynamic Data Driven Applications System*s, Ed. Erik Blasch, 2021.
2. L. Li and **Z. Li**, "Light Field Image Compression", Springer: *Dynamic Data Driven Applications System*s, Ed. Erik Blasch, 2018.
3. Z. Zheng, **Z. Li,** and A. Nagar, "Compact Deep Neural Networks for Image Understanding on Mobile Devices", Springer: *Mobile and Cloud Visual Media Computing*, Ed. G. Hua, and X. Hua, 2016.
4. H. Zheng, **Z. Li**, Y. Fu, and A. Katsaggelos, "Video Activity Recognition by Luminance Differential Trajectory and Aligned Projection Distance", *Handbook of Statistics,* Ed. V. Govindaraju et. al. Elsevier, 2013.

5. Y. Li, **Z. Li**, J. Huang, and M. Chiang, "Content-Aware Cross-Layer Optimizations in Video Networking", chapter in *Intelligent Multimedia Communication: Techniques and Applications*, Springer-Verlag, Ed. C.-W Chen, Z. Li, and S. Lian, 2010.
6. **Z. Li**, Y. Fu, J. Yuan, Y. Wu, A. K. Katsaggelos and T. S. Huang, "Multimedia Data Indexing", book chapter in *Semantic Mining Technologies for Multimedia Databases*, Ed. D. Tao, D. Xu, and X. Li, IGI Publishing, 2009.
7. **Z. Li**, A. K. Katsaggelos, and G. M. Schuster, "Rate-Distortion Optimal Video Summarization", book chapter in *Intelligent Multimedia Processing with Soft Computing*, pp. 171-204, editors: Y.P. Tan, K. H. Yap, and L. Wang, Springer-Verlag, Heidelberg, 2004.

Journal Papers in Review:

1. L. Li, **Z. Li**, S. Liu, H. Li, "Multiview-Video-Based Compression of Plenoptic Point Clouds", in review with *IEEE Transactions on Image Processing.*
2. D. Noor, **Z. Li**, S. Bhattacharyya, and G. York, "See SIT in Dark: Cascade Network for Learning DoG Pyramid in Dark", in review with *ACM Trans on Multimedia Computing, Communication, and Applications*, (TOMM), 2021.

Journal Papers: (underlined authors indicate supervised students/post-doc/visiting faculties)

3. W. Wu, Y. Liu and **Z. Li**, "Subband Differentiated Learning Network for Rain Streak Removal," in *IEEE Transactions on Circuits and Systems for Video Technology*, doi: 10.1109/TCSVT.2023.3246953, 2023.
4. W. Wu, H. Chang and **Z. Li**, "See SIFT in a Rain" in *IEEE Transactions on Circuits and Systems for Video Technology*, accepted, 2023.
5. Y. Zhang, G. Ding, D. Ding, Z. Ma and **Z. Li**, "On Content-aware Post-processing: Adapting Statistically Learned Models to Dynamic Content", in *ACM Trans on Multimedia Computing, Communication and Applications* (TOMM), 2023.
6. J. Wang, D. Ding, **Z. Li**, X. Feng, C. Cao and Z. Ma, "Sparse Tensor-based Multiscale Representation for Point Cloud Geometry Compression", accepted, *IEEE Trans. on Pattern Analysis & Machine Intelligence* (T-PAMI), 2023.
7. R. Liao, **Z. Li**, S. Bhattacharyya, and G. York, "PoseMapGait: A Gait Recognition Method based on Evolution of Pose Estimation Maps and GraphConvolutional Network", accepted, *Neurocomputing*, 2022.
8. L. Li, **Z. Li**, S. Liu, and H. Li, "Frame-level Rate Control for Geometry-based LiDAR Point Cloud Compression", accepted, *IEEE Trans. on Multimedia*, 2022.
9. Y. Sun, L. Li, **Z. Li**, S. Wang, S. Liu and G. Li, "Learning a Compact Spatial-Angular Representation for Light Field", *IEEE Trans. on Multimedia* (T-MM), 2022.
10. A. Akhtar, **Z. Li**, L. Li, G. Auwera, and J. Chen, "PU-Dense: Large Scale Photo-Realistic Point Cloud Upsampling", accepted with *IEEE Trans on Image Processing* (T-IP), 2022.
11. Y. Sun, **Z. Li**, S. Wang, and W. Gao, "Learning-based Depth-Guided Light Field Factorization for Compressive Light Field Display", *Optical Express*, 2022.
12. **Z. Li**, S. Liu, F. Dufaux, L. Li, G. Li and C. . -C. J. Kuo, "Guest Editorial Introduction to the Special Issue on Recent Advances in Point Cloud Processing and Compression," in *IEEE Transactions on Circuits and Systems for Video Technology*(T-CSVT), vol. 31, no. 12, 2021, doi: 10.1109/TCSVT.2021.3129071.
13. D. Yang, Y.-X Zou, **Z. Li**, and G. Li, "Learning Human-Object Interaction via Interactive Semantic Reasoning", *IEEE Trans on Image Processing* (T-IP), doi: 10.1109/TIP.2021.3125258, 2021.
14. L.Li, **Z. Li**, S. Liu and H. Li, "Motion Estimation and Coding Structure for Inter-prediction of LiDAR Point Cloud Geometry", accepted, *IEEE Trans on Multimedia* (T-MM), 2021.
15. Y. Mei, L. Li, **Z. Li**, and F. Li, "Learning-Based Scalable Image Compression with Latent-Feature Reuse and Prediction", accepted, *IEEE Trans on Multimedia* (T-MM), 2021.
16. R. Hang, Q. Liu, and **Z. Li**, "Spectral Super-Resolution Network Guided by Intrinsic Properties of Hyperspectral Imagery", *IEEE Trans on Image Processing*(T-IP) doi:10.1109/TIP.2021.3104177, 2021.
17. W. Jia, L.Li, **Z. Li**, and S. Liu, "Deep Learning Geometry Compression Artifacts Revomal for Video Based Point Cloud Compression", accepted with *Int'l Journal on Computer Vision* (IJCV), 2021.
18. A. Akhtar, L. Li, W. Gao, **Z. Li**, W. Jia and S. Liu, "Video-based Point Cloud Compression Artifact Removal", accepted, *IEEE Trans on Multimedia* (T-MM), 2021.
19. X. Sheng, L. Li, D. Liu, Z. Xiong, **Z. Li**, and F. Wu, "Deep-PCAC: An End-to-End Deep Lossy Compression

Framework for Point Cloud Attributes", accepted, *IEEE Trans on Multimedia* (T-MM), 2021.

20. W. Jia, L. Li, **Z. Li** and S. Liu, "Convolutional Neural Network-based Occupancy Map Accuracy Improvement for Video-based Point Cloud Compression", accepted, *IEEE Trans on Multimedia* (T-MM), 2021.
21. R. Liao, W. An, **Z. Li**, and S. S. Bhattacharyya, "A novel view synthesis approach based on view spacecovering for Gait Recognition", accepted, *Neurocomputing* , 2021.
22. W. Ji, T. Ebrahimi, **Z. Li**, J. Yuan, D. Wu, Y. Xin, "Guest Editorial: Emerging Visual IoT Technologies for Future Communications and Networks", *IEEE Wireless Comm.* , vol. 28(4): 10-11 (2021).
23. W. Jia, L. Li, **Z. Li**, X. Zhang and S. Liu, "Residual Guided In-Loop Filter for HEVC", accepted, *ACM Trans on Multimedia Computing, Communication and Applications*(TOMM), 2021.
24. T. Li, K. Zhang, S. Shen, B. Liu, Q. Liu, and **Z. Li,** "Image Co-saliency Detection and Instance Co-segmentation using Attention Graph Clustering based Graph Convolutional Network", accepted, *IEEE Trans on Multimedia* (TMM), 2021.
25. Q. Yang, Z. Ma, Y. Xu, **Z. Li**, and J. Sun, "Inferring Point Cloud Quality via Graph Similarity", to appear, *IEEE Trans on Pattern Analysis & Machine Intelligence*, (T-PAMI), 2020.
26. Y. Li, Y. Yi, D. Liu, L. Li, **Z. Li**, and H. Li, "Neural Network Based Cross-Channel Intra Prediction", *ACM Trans on Multimedia Computing Communication and Applications* (TOMM), 2020.
27. W. Jia, L. Li, **Z. Li**, S. Zhao and S. Liu, Scalable Hash From Triplet Loss Feature Aggregation for Video De-Duplication, *Journal of Visual Communication & Image Representation* (JVCIR), 2020.
28. L. Li, **Z. Li**, S. Liu, H. Li, "Efficient Projected Frame Padding for Video-based Point Cloud Compression", accepted, *IEEE Transactions on Multimedia,*(T-MM) 2020.
29. L. Li, **Z. Li**, S. Liu, H. Li, "Rate Control for Video-based Point Cloud Compression", accepted with *IEEE Transactions on Image Processing* (T-IP)*,* 2020.
30. H. Zhang, L. Song, L. Li, **Z. Li**, and X.K. Yang, "Compression Priors Assisted Convolutional Neural Network for Fractional Interpolation", to appear, *IEEE Transactions on Circuits and Systems for Video Technology.*(T-CSVT)
31. R. Hang, **Z. Li**, Q. Liu, P. Ghamisi and S. Bhattacharyya, "Hyperspectral Image Classification with Attention Aided CNNs", to appear, *IEEE Trans. on Geoscience & Remote Sensing*(T-GRS), 2020.
32. W. Zhu, Z. Ma, Y. Xu, L. Li, and **Z. Li**, "View-Dependent Dynamic Point Cloud Compression", *IEEE Transactions on Circuits and Systems for Video Technology* (T-CSVT), 2020.
33. Z. Zhang, X. Zhao, X. Li, L. Li, S. Liu, and **Z. Li** "Fast DST-7/DCT-8 with Dual ImplementationSupport for Versatile Video Coding",accepted, *IEEE Trans on Circuits & Sys. for Video Tech* (T-CSVT), 2020.
34. R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia and Q. Liu, "Classification of Hyperspectral and LiDAR Data Using Coupled CNNs," in *IEEE Transactions on Geoscience and Remote Sensing*, (T-GRS), vol. 58, no. 7, pp. 4939-4950, July 2020, doi: 10.1109/TGRS.2020.2969024.
35. Li Li, N. Yan, **Z. Li**, S. Liu, H. Li, "λ-domain Perceptual Rate Control for 360-degree Video Compression", accepted with *IEEE Journal of Selected Topics in Signal Processing*,(JSTSP) 2020.
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